Name: Charandeep N C
Reg no: 192472196
Slot: D
Course Name: Mobile Computing

EXP 3

Develop a Flowchart in Figma Depicting the Operation of a Network Switching Subsystem (NSS)

Aim

To develop a flowchart using Figma that illustrates the operation of the Network Switching Subsystem (NSS) in a mobile cellular communication system

Procedure

1. Open Figma and create a new design file.
2. Create a flowchart workspace using a suitable frame.
3. Add a **Start** symbol to indicate the beginning of the process.
4. Add a process block representing a mobile station initiating a call through the Base Transceiver Station (BTS) and Base Station Controller (BSC).
5. Add a process block for the Mobile Switching Centre (MSC), which processes the call request.
6. Include the Home Location Register (HLR) and Visitor Location Register (VLR) to retrieve subscriber and location information.
7. Add the Authentication Centre (AuC) to verify subscriber identity when required.
8. Add decision blocks to check subscriber authentication and destination availability.
9. Connect the blocks using arrows to represent call routing and decision paths.
10. Add a process block for establishing the connection between the calling and receiving mobile stations.
11. Include an **End** symbol to indicate call termination and release of network resources.
12. Review the flowchart and arrange the symbols neatly for clarity.

Implementation



Figma Link: https://www.figma.com/design/U4EG6pfQDYzQ84xLfUMgT1/EXP-3?m=auto&t=oC7SqyCm2IYS8ANn-6

Result

Thus, a flowchart representing the operation of the Network Switching Subsystem in a mobile cellular system was developed using Figma, illustrating subscriber authentication, location information retrieval, call routing, connection establishment, and call termination.